

Payment Method Choice at Checkout

Homework Assignment No. 2

Matias Arriagada R.

May 2026

Universidad de Concepción

Introduction & Phenomenon Identification

Payment Method Choice at Checkout

The Decision Context

At the checkout screen of an e-commerce platform, consumers make a discrete choice among several mutually exclusive payment methods.

Why Study Payment Method Choice?

- **For Merchants:** Transaction fees vary (e.g., Credit Card 2% vs. UPI 0%). Merchant can steer choices.
- **For Fintechs & Banks:** Understand what features (speed, safety, fees) attract users.
- **For Checkout Design:** Reduce cart abandonment by minimizing friction.

Econometric Modeling

- Matches utility maximization theory: $U_{ij} = V_{ij} + \epsilon_{ij}$.
- Incorporates alternative-specific attributes and individual characteristics.

Data Selection & Source

Relational Datasets merged on `customer_id`

1. Orders Dataset (`orders.csv`)

- **25,000 transactions**
- Contains choice variable (`payment_method`), transaction-level factors (`total_amount_usd`), and web metrics (`session_duration_minutes`).

2. Customers Dataset (`customers.csv`)

- **8,000 unique customers**
- Demographics (age, gender, country) and loyalty info (`membership_tier`).

Choice Set & Attributes Definition

- **Top 4 Alternatives:** Credit Card, Debit Card, PayPal, UPI / Digital Wallet.
- **Calculated fee (`fee_usd`):** Credit Card (2.0%), Debit Card (0.5%), PayPal (3.0%), UPI (0.0%).
- **Assumed speed (processing time):** UPI (1.5s) < Credit (2.0s) < Debit (2.5s)

Exploratory Data Analysis

Sample Demographics and Market Shares

Choice Distribution (N = 19,880)

- **Credit Card:** 43.04%
- **Debit Card:** 24.93%
- **PayPal:** 20.54%
- **UPI / Wallet:** 11.48%

High Demographic Homogeneity

- **Age:** Average age is ~35.5 years across all payment choices.
- **Gender:** Female percentage is ~48-50% in all groups.
- **Order Value:** Mean order is ~\$126 across all groups.

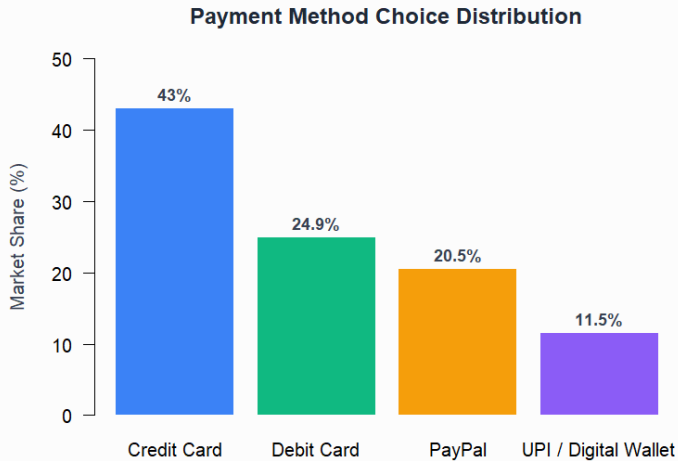
Table 1: Averages by Choice Group

Method	Mean Age	Pct Female	Mean Order
Credit Card	35.41	48.7%	\$126.26
Debit Card	35.67	48.2%	\$127.34
PayPal	35.56	50.4%	\$124.90
UPI / Wallet	35.67	49.1%	\$127.15

- Demographics and order values do not show strong direct correlations with choices, pointing to a balanced/randomized distribution.

Data Visualization: Market Shares

Distribution of Payment Choices



Data Visualization: Demographics & Countries

Choice by Membership Tier & Top Countries

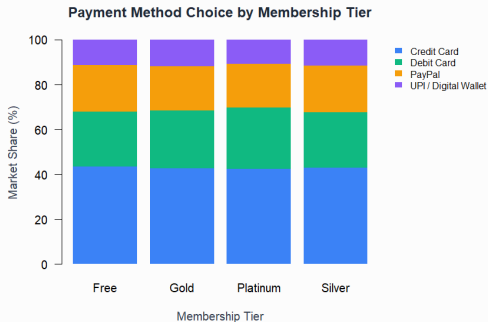


Figure 2: Market Share by Membership Tier

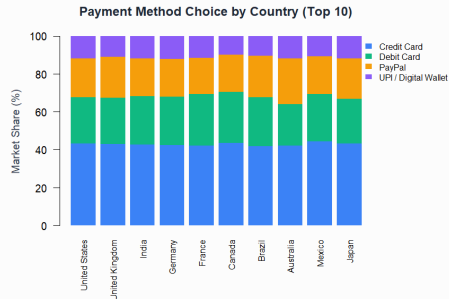


Figure 3: Market Share by Top 10 Countries

Data Selection & Filters

Establishing the Estimation Sample

Why Filters are Necessary

To ensure our sample matches the econometric definition of a finalized checkout and a well-defined choice set.

1. Order Status Filter (Delivered/Returned)

- Excluded Cancelled (1,456) and Processing (1,027) orders since checkouts were not completed.
- *Sample size:* 25,000 $\rightarrow$ 22,517.

2. Choice Set Filter (Top 4 Payment Methods)

- Excluded Bank Transfer (832), BNPL (1,358), and Crypto (447) due to low volumes ($<12\%$ combined).
- *Sample size:* 22,517 $\rightarrow$ **19,880 transactions** (Final Sample).

Baseline Econometric Models

Multinomial Logit (MNL) Models

Model 1: Attribute-Only Model (No ASCs)

$$U_{ij} = \beta_{price} \cdot fee_usd_{ij} + \beta_{time} \cdot processing_time_j + \beta_{quality} \cdot security_score_j + \gamma_j \cdot X_i + \epsilon_{ij}$$

- Isolates speed, cost, and safety attributes. UPI is used as the base.

Model 2: Full Intercept Model (With ASCs)

$$U_{ij} = ASC_j + \beta_{price} \cdot fee_usd_{ij} + \gamma_j \cdot X_i + \epsilon_{ij}$$

- Captures unobserved alternative-specific preferences. Credit Card is the base ($ASC = 0$).

Estimation Results

Model Comparison Summary

Variable	Model 1 (Attribute-Only)		Model 2 (With ASCs)	
	Estimate	p-value	Estimate	p-value
$ASC_{DebitCard}$	—	—	-0.6674	< 2e-16 ***
ASC_{PayPal}	—	—	-0.7311	< 2e-16 ***
ASC_{UPI}	—	—	-1.4153	< 2e-16 ***
Fee (USD) (β_{price})	0.0023	0.618	-0.0038	0.420
Time (s) (β_{time})	-0.2273	< 2e-11 ***	—	—
Security ($\beta_{quality}$)	0.4182	< 2e-14 ***	—	—
Log-Likelihood	-25,625.91		-25,492.30	
AIC / BIC	51,293.82 / 51,459.67		51,028.60 / 51,202.34	
McFadden R^2	0.0000		0.0004	

- **Time** is negative and highly significant.
- **Security** is positive and highly significant.
- **Fee** is statistically insignificant in both models.

Willingness-to-Pay (WTP)

Econometric Interpretation and Volatility

Willingness-to-Pay Estimates (Model 1)

- **WTP for Speed:** $-\$98.51$ per second saved (implied sign-flip).
- **WTP for Security:** $\$181.23$ per unit of security increase.

Methodological Insights: Why are these values unstable?

- The price coefficient ($\beta_{price} = 0.0023$) is statistically indistinguishable from zero ($p = 0.618$).
- In synthetic data, payment choices were generated independently of the tiny transaction fees.
- When the cost coefficient is close to zero, dividing any other coefficient by it leads to extremely large and volatile values.
- **Key Takeaway:** A cost attribute must have a robustly identified, statistically significant negative coefficient to serve as a reliable numeraire for converting utility into monetary value.

Conclusions & Future Research

Summary of Findings

1. Successful Setup of Choice Framework

- Merged relational data and defined an estimation sample of 19,880 clean checkouts.
- Defended decision to keep observations with missing customer ratings.

2. Confirmed Attribute Preferences

- Checkout speed and payment security are powerful drivers of consumer utility.

3. Econometric Insight on Cost Identification

- Uncovered that weakly identified price coefficients lead to unusable WTP estimations, highlighting a common hazard in discrete choice modeling.

4. Next Steps for Homework 3

- Introduce Nested Logit models (grouping Cards vs. Digital/Third Party methods).
- Explore interaction terms to capture segment-specific cost sensitivities.